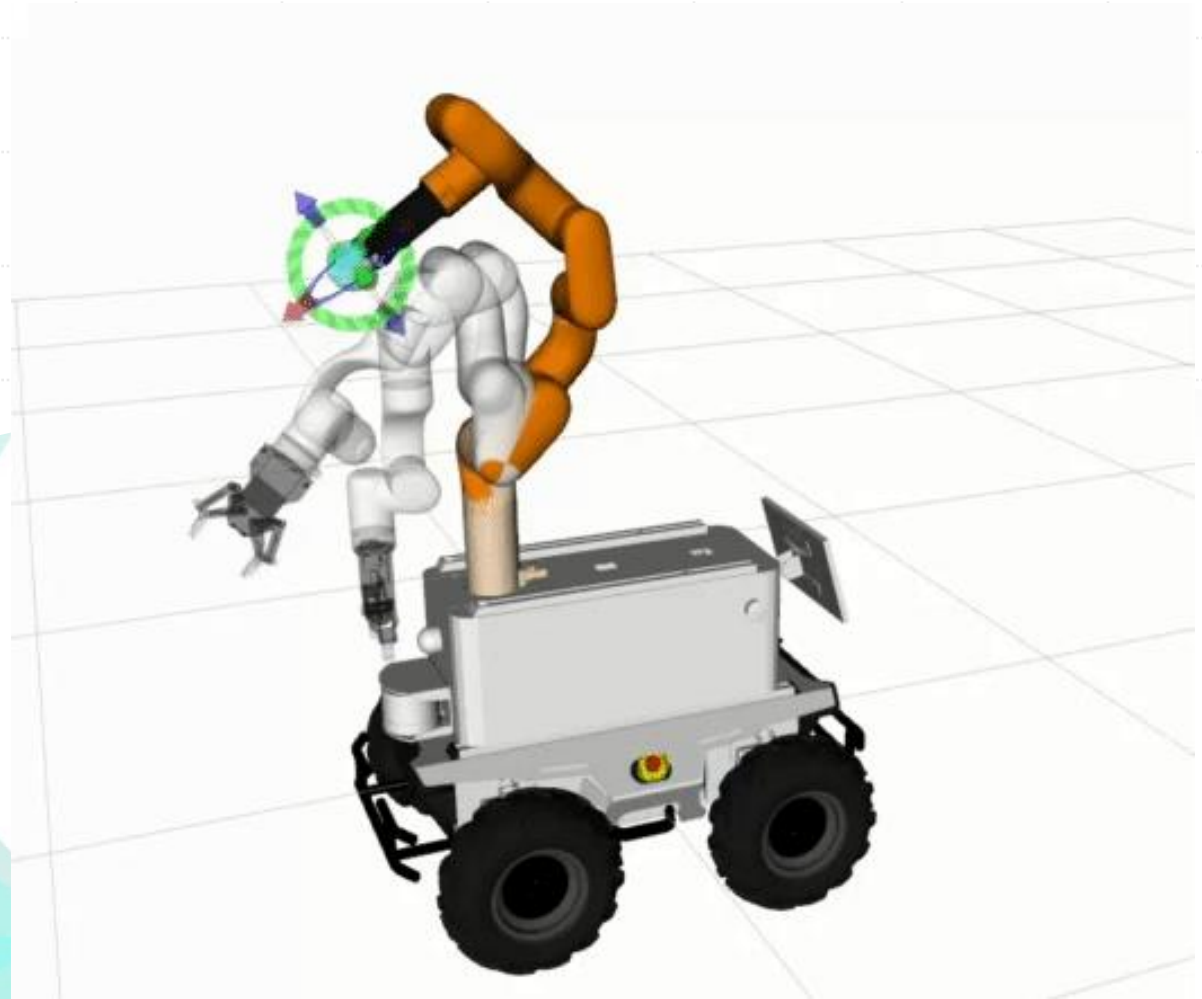


Object Manipulation For Unseen Objects

By:

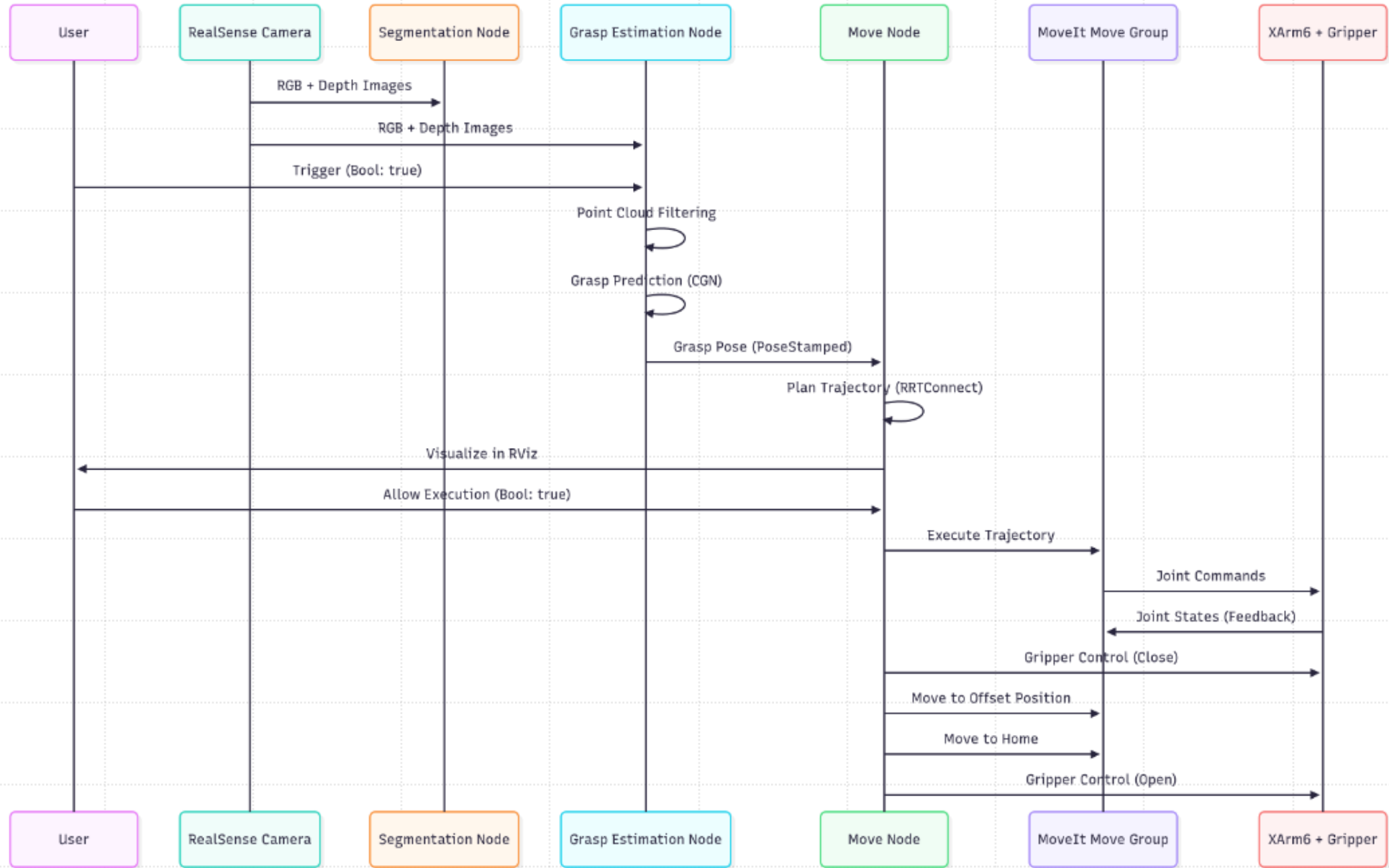
- Jamin Rahman Jim
- Muhammad Faran Akram
- Pravin Oli
- Sherif Samy



Motivation

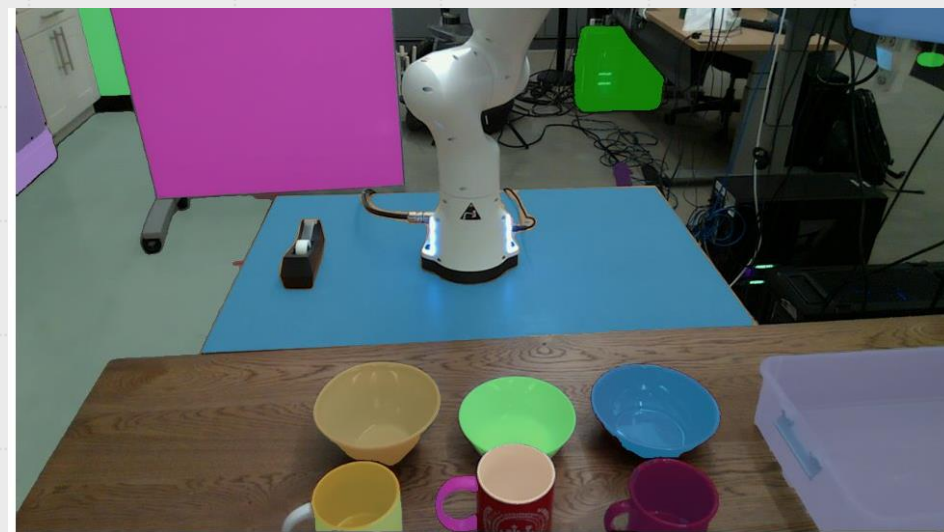
- Industrial manipulators struggle with objects of diverse shapes, sizes, and materials.
- Traditional grasping depends on predefined poses, limiting adaptability to unseen objects.
- Deep learning (RL/IL) methods improve flexibility but demand costly training, accurate simulation, and large datasets.
- A robust, adaptive, and data-efficient grasping strategy is needed for real-world manipulation.
- Point-cloud-based 6-DoF grasp prediction offers a practical, generalizable solution for unseen objects.

System Overview



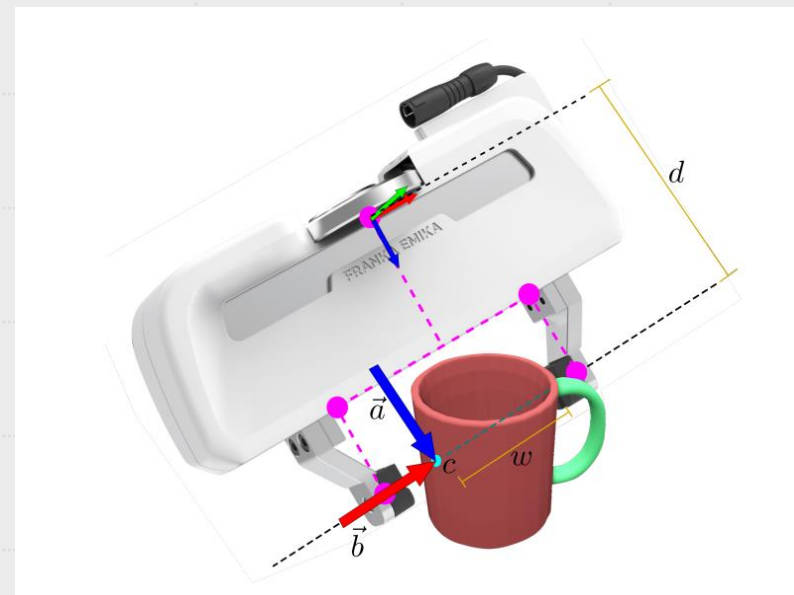
Instance Segmentation

- UOIS-Net
 1. Predict offsets to PCD coordinates to cluster them.
 2. Cluster points using Gaussian Mean-Shift
 3. Refine each local cluster mask separately
- Segment Anything Model (SAM)
 1. ViT Image Encoder
 2. Prompt Encoder
 3. Mask Decoder

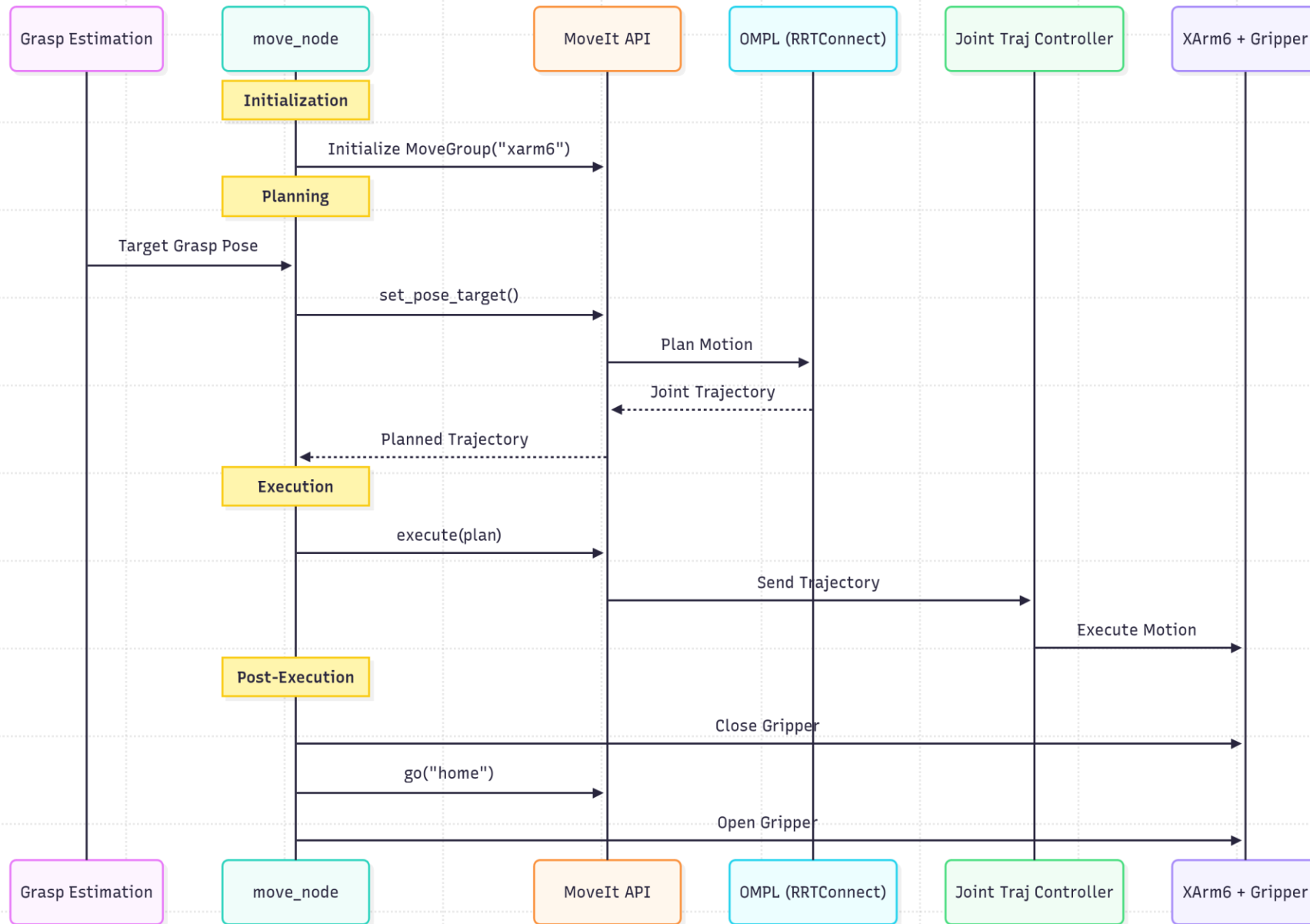


Grasp Pose Prediction

- Contact GraspNet
 - Processes input PCD using the PointNet++ Architecture
 - Produces feature vectors for a subset of points
 - Uses three output heads
- GraspNet Outputs
 - Grasp confidence score $[0, 1]$
 - Grasp width (10 discrete bins)
 - Approach and grasp baseline vectors (processed through Gram-Schmidt orthonormalization)



ROS_Control & Moveit









Thank You